Towards Comprehensive Legal Document Analysis: A Multi-Round RAG Approach

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Abstract

Legal document review is a time-consuming and highly specialized task, and the capabilities of intelligent legal review systems are limited and insufficient to complete detailed reviews. Traditional methods struggle with cross-references, dependencies, and context-dependent clauses. Our work introduces a multi-round RAG framework for legal document analysis, which iteratively refines queries and aggregates context to improve recall and understanding. Experiments on diverse contracts show a recall of 78.67%, outperforming baseline (57.33%) and single-round RAG (74.67%). Our analysis shows that iterative refinement effectively filters irrelevant results despite reduced precision. The multi-round approach halves missed cross-clause dependencies but reveals limitations in numerical consistency and obligation scope detection. These insights advance RAG for legal applications and provide a foundation for future work on scalable and accurate contract review.

CCS Concepts

• **Computing methodologies** → **Information extraction**; • **Information systems** → **Language models**; **Task models**.

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ICMR '25, Chicago, IL, USA.
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ACM ISBN 979-8-4007-1592-1/25/07
https://doi.org/10.1145/3731715.3733451

Keywords

Retrieval-Augmented Generation, Multi-Round Retrieval, Document Retrieval, Contract Review, Large Language Model

ACM Reference Format:
Wutong Zhang, Hefeng Zhou, Qiang Zhou, Yunshen Li, Yuxin Liu, Jiong Lou, Chentao Wu, and Jie Li. 2025. Towards Comprehensive Legal Document Analysis: A Multi-Round RAG Approach. In *Proceedings of the 48th International ACM ICMR Conference on Research and Development in Information Retrieval (ICMR '25), June 30–July 3, 2025, Chicago, IL, USA.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3731715.3733451

1 Introduction

The world of legal documentation has undergone a dramatic shift in recent years as organizations grapple with an overwhelming surge in both the volume and complexity of contracts due to increasing globalization and stricter regulatory requirements. This transformation has made traditional manual review processes increasingly expensive and error-prone. Legal professionals now routinely deal with documents filled with intricate cross-references, complex dependencies, and specialized terminology, where even a single oversight can lead to significant financial losses, legal disputes, or compliance violations [13, 14, 18]. The stakes are especially high in heavily regulated industries like finance, healthcare, and government procurement, where compliance demands exceptional precision and thoroughness [28].

Despite significant advancements in artificial intelligence (AI), particularly with the rise of Large Language Models (LLMs), existing approaches still fall short when applied to the specialized demands of legal contexts [7, 10]. Traditional rule-based systems and keyword-based retrieval methods have proven inadequate for handling the semantic complexity of legal texts, often resulting in low accuracy and poor recall for clauses that depend on context [4, 8, 16]. Even advanced neural retrieval models, such as dense passage retrieval (DPR), struggle to generalize across the diverse structures and terminologies found in legal documents [19, 27].

A significant limitation of current retrieval methods is their reliance on single-pass retrieval, which fails to capture the nuanced relationships between clauses scattered throughout a contract. Legal documents often require multiple iterations of retrieval and

reasoning to ensure that interdependent provisions are correctly interpreted in context. Without an iterative approach, crucial dependencies may be overlooked, resulting in misinterpretation of obligations, financial terms, or compliance risks.

Three major challenges stand out when applying AI to legal document analysis. First, retrieval accuracy must be fine-tuned for legal texts, where understanding precise terminology and domain-specific semantics is critical [18, 19]. Second, resolving cross-references and dependencies across multiple sections of a contract requires sophisticated mechanisms that go beyond simple sequential reading, especially when clauses are scattered throughout the document [7, 26]. Third, verifying numerical constraints—such as payment terms, deadlines, and performance metrics—presents unique computational challenges that demand robust context tracking and arithmetic consistency checks [10, 21].

In response to these challenges, there is growing interest in retrieval-augmented generation (RAG) systems, which combine retrieval mechanisms with the generative capabilities of LLMs to produce more accurate and context-aware outputs [4, 11]. Recent studies have highlighted the potential of iterative multi-round retrieval frameworks to improve cross-clause understanding and information extraction in knowledge-intensive domains significantly [15, 17, 27]. By incorporating an iterative refinement process, multi-round retrieval addresses the limitations of traditional single-round retrieval by dynamically expanding the retrieved context until a more complete and accurate legal analysis is achieved. Building on the IM-RAG framework, which iteratively refines queries and retrieves additional context until a comprehensive understanding is achieved, our approach aims to tackle the specific challenges posed by legal documents. By leveraging multi-round retrieval, our model improves accuracy in clause interpretation and cross-clause reasoning, effectively handling interdependent legal provisions that traditional retrieval systems struggle with.

In this research, we introduce a multi-round RAG framework tailored for legal document analysis, focusing on progressive context refinement and enhanced retrieval. The framework is designed to iteratively build understanding across multiple rounds, enabling deeper cross-clause analysis and improved numerical consistency verification. By incorporating semantic clause-level retrieval and iterative query refinement, our system aims to reduce missed dependencies while maintaining a balance between recall and precision—a critical requirement for high-stakes applications like contract review and compliance verification [7, 15, 27].

We conduct an in-depth error analysis, categorizing common challenges in automated contract analysis and offering practical insights for future research. This research advances the theoretical understanding of RAG in legal contexts and offers practical tools and methodologies for scaling these solutions in real-world legal environments [22, 25, 28].

By addressing these core challenges in automated legal document analysis, our work bridges the gap between the capabilities of advanced language models and the practical needs of legal professionals. The proposed multi-round retrieval framework lays the groundwork for next-generation intelligent legal technology solutions, offering greater accuracy, reliability, and scalability in contract review workflows. This is particularly crucial in industries where even a minor oversight can have significant legal and financial consequences [10, 13, 28].

2 Preliminary Study

Verifying compliance with legal documents is challenging due to complex legal terminology, structural dependencies, and the dynamic regulatory landscape. Both manual and automated methods have significant shortcomings [2]. While Retrieval-Augmented Generation (RAG) models offer a promising way to enhance the precision and effectiveness of compliance verification, legal documents' specialized terminology, nested clauses, and conditional statements pose major challenges for automated processing. Traditional NLP systems often fail to accurately interpret such complexity, leading to potential misinterpretations in compliance verification, and existing approaches frequently rely on limited context, causing misunderstandings of legal texts. Moreover, legal documents typically contain extensive cross-references and interdependencies. Compliance obligations often span multiple sections with scattered amendments and exceptions, necessitating advanced parsing systems to maintain logical connections and ensure document-wide consistency. Meanwhile, the dynamic regulatory environment demands that compliance verification systems adapt to new standards while retaining backward compatibility. Current compliance verification techniques are flawed. Manual verification is resource-intensive and inconsistent, especially for complex documents, while automated keyword-matching techniques, though faster, fail to capture semantic nuances, leading to higher false positive rates and difficulty in identifying implicit requirements. Conventional NLP systems for general language processing tasks demonstrate limited effectiveness in legal settings. There are some substantial shortcomings, such as the inability to handle nested logical conditions, interpret temporal references, incorporate numerical constraints, and address cross-document dependencies

While Large Language Models (LLMs) demonstrate impressive natural language understanding capabilities, they often produce outputs that lack the precision required for legal applications. The hallucination phenomenon, where models generate plausible but factually incorrect information, poses particular risks in compliance contexts where accuracy is paramount. Hassani et al. [5] discuss instances where LLMs generate plausible but incorrect or nonsensical information, posing risks in legal applications.

RAG models integrate retrieval and generation to enhance precision by using credible legal references, thus addressing regulatory changes and reducing computational costs. This efficiency supports scalable compliance verification. In a comparison with a Baseline LLM model, our Multi-round RAG framework better identifies issues in interconnected contract clauses (Figure 1) by capturing dependencies missed by single-pass models.

To illustrate the above challenges in detail, we conducted tests using existing state-of-the-art LLMs. In this real-world example:

- Clause **4.3.2** specifies that Party B must submit regular service reports. Party A reserves the right to withhold payments and confiscate part of the performance bond if the assessment is unsatisfactory.

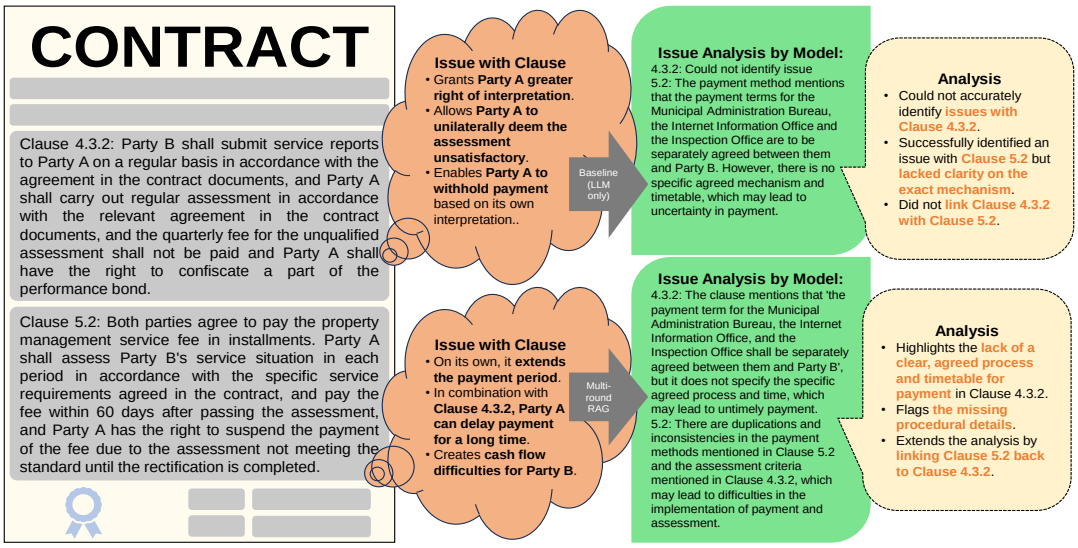


Figure 1: Comparative analysis highlights the identification performed by basic model and our method on Clauses 4.3.2 and 5.2.

- Clause **5.2** further allows Party A to delay payments until assessments meet specific service requirements, yet it lacks a clearly defined process and timetable for such assessments.

The Basic model identifies isolated issues in Clause 5.2 but fails to connect it with Clause 4.3.2, thus overlooking the unilateral power it grants Party A to indefinitely delay payments, potentially causing cash flow difficulties for Party B. In contrast, the Multi-round RAG framework iteratively enhances its query by retrieving relevant context from multiple clauses. It highlights potential payment and assessment issues by identifying missing procedural details in Clause 4.3.2 and linking them to Clause 5.2. This example underscores the necessity of multi-round retrieval for capturing cross-clause dependencies often overlooked by single-pass models, thereby enhancing contract review accuracy. The iterative process improves context understanding, resolves cross-references, constructs comprehensive context models, and maintains logical consistency across complex compliance requirements, while reliable source-based outputs reduce the risk of inaccuracies. RAG models enhance accuracy in legal contexts by sourcing reliable data, as Hassani et al. highlight the role of retrieval components in minimizing inaccuracies. In summary, while traditional methods and LLMs have contributed to legal compliance verification, RAG models present a more advantageous solution by effectively combining retrieval and generation capabilities. This hybrid approach addresses the limitations of existing solutions, offering enhanced accuracy, efficiency, and adaptability in managing the complexities of legal document compliance.

3 Related Works

3.1 Document Understanding and Retrieval

Document retrieval systems have evolved from traditional keyword-based methods to advanced neural models, significantly enhancing the accuracy and relevance of search results [29]. As highlighted by Kenter et al. [8], early retrieval methods relied primarily on lexical matching and term frequency-based ranking algorithms. While these approaches were effective for simple retrieval tasks, they often fell short when dealing with context-dependent queries, limiting their utility in complex domains such as legal analysis. Dense Passage Retrieval (DPR) introduced by Karpukhin [6] represents a significant step forward in document retrieval, addressing the limitations of traditional sparse vector-based methods such as BM25 [23]. Unlike BM25, which relies on exact keyword matching, DPR uses dense vector representations learned through a dual-encoder framework, allowing it to match semantically similar phrases even if they share no common tokens.

Bidirectional Encoder Representations from Transformers (BERT) [3] is considered a groundbreaking work in natural language processing in the era of artificial intelligence. It significantly enhances language understanding capabilities by introducing bidirectional context encoding and provides a robust foundation model for various natural language processing tasks. Further progress was made with introducing the ColBERT model [9], which integrates BERT-based contextual embeddings with efficient late interaction mechanisms. While this approach balances effectiveness and computational efficiency for complex document retrieval tasks, ColBERT may still struggle to capture the intricate relationships in legal texts. Recent efforts have increasingly centered on enhancing the efficiency and scalability of neural retrieval models to overcome existing limitations. For example, Bruch et al. [1] introduced ReNeuIR, a framework designed to optimize neural retrieval systems for large-scale document processing. Although ReNeuIR demonstrates significant improvements in scalability, its application to legal documents requires particular attention to the distinct linguistic and structural characteristics defining legal texts. These ongoing developments underscore the necessity of tailoring retrieval systems to

the unique demands of legal analysis, ensuring both accuracy and practical feasibility in real-world settings.

3.2 Legal Document Analysis

Legal document analysis is inherently complex due to the sophisticated language, intricate logical constructs, and specialized terminology that characterize legal texts. Traditional rule-based systems have long been employed for tasks such as legal text classification and clause extraction. While these systems often prove effective for well-structured legal documents, they tend to lack the flexibility needed to handle unstructured or less standardized formats, ultimately limiting their practicality in real-world applications. The advent of large language models (LLMs) has revolutionized legal document analysis by addressing these limitations. For example, Limsopatham [12] demonstrated the potential of fine-tuning BERT for legal text classification, achieving significantly higher accuracy than conventional methods. Beyond classification, LLMs have been successfully applied to tasks like legal text summarization and compliance verification, automating processes such as identifying non-standard contract clauses. However, fine-tuning LLMs for legal texts also presents challenges, including the need for extensive domain-specific training data and the risk of overfitting to specific legal contexts, which can hinder generalization across diverse legal documents.

Resolving cross-references and dependencies among clauses is one of the primary difficulties in legal document analysis[20]. Since contracts often contain interconnected sections, each must be interpreted in relation to the others to fully capture their meaning. Meeting this challenge requires systems adept at retrieving relevant clauses and understanding the relationships and contextual implications between them. In our work, we strive to develop models that can skillfully navigate and interpret these intricate document structures, leading to a more thorough and precise understanding of legal content.

3.3 RAG System

Retrieval-Augmented Generation (RAG) systems have emerged as a powerful approach to improving document understanding by combining retrieval mechanisms with the generative capabilities of language models. Wang et al. [24] introduced M-RAG, a framework that segments the retrieval process to enhance context understanding and generation quality, particularly for large documents where pinpointing specific contextual information is crucial. Building on this, Yang et al. [27] proposed IM-RAG, which integrates information retrieval systems with large language models (LLMs) to enable multi-round retrieval through "learning inner monologues." In this framework, the LLM acts as the central reasoning model, iteratively generating queries to gather information via the retriever or delivering final answers based on conversational context. Adding a refiner component further enhances communication between the reasoning model and the IR modules, improving the overall effectiveness of multi-round retrieval. However, applying these frameworks to legal contract analysis requires modifications to accommodate the specialized language and intricate structures typical of legal texts. Our method extends IM-RAG by introducing a contract-specific iterative retrieval module that aligns legal provisions, mitigating interpretation errors arising from isolated clause retrieval.

Recent advancements in RAG systems have focused on refining multi-task retrieval and parameter-efficient fine-tuning. Salemi and Zamani [17] proposed uRAG, a unified ranking framework that optimizes retrieval across multiple RAG systems, improving scalability and performance through shared optimization across diverse downstream tasks such as question answering, fact verification, and relation extraction. Similarly, Rangan and Yin [15] introduced a fine-tuning enhanced RAG system incorporating Quantized Influence Measure (QIM) as an "AI Judge" to prioritize highly relevant retrieval results. This innovation enhances precision and memory efficiency using QLoRA for parameter-efficient fine-tuning, significantly improving the system's retrieval accuracy and adaptability. Integrating these RAG systems into legal document analysis presents a promising solution to the limitations of traditional methods. By merging the retrieval strengths of information retrieval systems with the generative power of LLMs, RAG systems can significantly improve context understanding and the quality of generated responses, which is particularly valuable in the legal domain where documents are often lengthy, complex, and filled with intricate interdependencies that require careful analysis.

4 Methods

4.1 System Overview

The core of this paper is the implementation of the retrieval-augmented generation (RAG) framework explicitly designed for the analysis and review of legal documents. Unlike the traditional single-round systems, this framework adopts the multi-round approach to query generation and output refinement, allowing the system to build a more complete and comprehensive understanding of the document after multiple rounds of inference. This multi-round strategy follows the human thinking process by revisiting different contract sections to clarify and confirm details, assuring a more accurate and thorough analysis.

The System Architecture Diagram (Figure 2) illustrates the end-to-end process of the multi-round RAG framework. The process begins when the user attaches the contract to be analyzed. Rather than asking users to craft queries or manually change parameters, the system relies on a predefined prompt designed explicitly for legal document analysis. This prompt identifies the model as a legal specialist, directing it to examine each contract section and generate relevant insights. At the same time, a standardized output format—often JSON—ensures the analysis follows a consistent structure. Within this framework, the model splits each contract into smaller, manageable segments, each capped at 512 tokens to maintain semantic coherence. These segments are then processed using the M3E model, which is purpose-built to capture the subtleties of legal language. Next, the system determines the segments' importance by evaluating their semantic similarity to the query, where the contract's content and the predefined prompt collectively serve as this implicit query. Finally, the segments with the highest relevance are passed to the reasoning phase for deeper examination.

After the document is divided into chunks and pertinent sections are obtained, the LLM Reasoning module is essential for interpreting and analyzing the gathered chunks. The module examines and

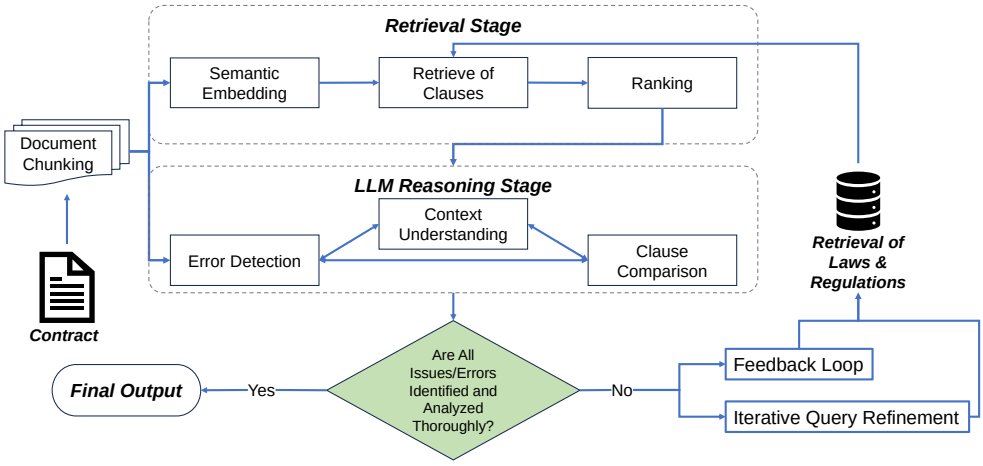


Figure 2: Framework: An iterative process that starts with document chunking and clause retrieval, proceeds with LLM-based reasoning for detection and clause analysis, and enters refinement cycles for re-retrieval and re-evaluation if issues persist, ensuring comprehensive legal analysis until all errors are resolved to deliver an accurate and legally aligned contract review.

evaluates every clause to identify contradictions, logical inconsistencies, and overlapping conditions. It enhances comprehension by iteratively refining its understanding, filling in missing context, and resolving ambiguities, especially in numerical terms and cross-referenced clauses. Additionally, in the initial retrieval phase, the model conducts error detection, recognizing numerical discrepancies, cross-clause interdependencies, and contractual disparities that may signify compliance risks or conflicts. Should context gaps or ambiguous clauses be identified, the system will revert to the retrieval stage to collect supplementary context for enhancement. After the initial round of analysis, the system proceeds to retrieve relevant laws and regulations from the RAG dataset, grounding the contract analysis in legal frameworks. This ensures that any detected errors or ambiguities are checked against authoritative legal references. The model refines the analysis based on this new legal context, adjusting queries to capture additional clauses or relationships as needed. The iterative retrieval process continues, with the system refining its queries in each round based on the context of previously retrieved information and legal references. This process ensures that the clauses align with the laws and regulations and checks for cross-clause conflicts or misalignments. The system can identify issues like contradictory terms, incomplete references, or inconsistent obligations by comparing related clauses. The iterative nature of this retrieval process is designed to gradually improve the accuracy and completeness of the legal review, helping ensure that all relevant clauses are considered in the context of the entire document.

In this multi-round process, the final output is generated by marginalizing the retrieved clauses at each round. Formally, the probability of generating the final answer y. given the contract x, is represented as:

$$p(y|x) = \sum_{z_1 \in \text{top-K}(p(\rho|q_1))} p_n(z_1|x) \cdot p(y|x, z_1) \cdot \sum_{z_2 \in \text{top-K}(p(\rho|q_2))} p_n(z_2|x) \cdot p(y|x, z_2) \cdots$$

where $p(y \mid x)$ represent the probability of generating the final legal assessment based on the contract x. The term $p_n(z_1 \mid x)$ denotes the probability of retrieving a clause z_1 in the first retrieval round, which is determined based on its semantic similarity to the initial query q_1. The probability $p(y \mid x, z_1)$ corresponds to the likelihood of generating the answer given the first retrieved clause. As retrieval progresses, the query is refined, leading to a second retrieval round in which a new set of clauses z_2 is retrieved with probability $p_n(z_2 \mid x)$, conditioned on the updated query q_2. The summation over the top-K retrievals in each round ensures that the final answer is computed by considering multiple relevant clauses from different retrieval iterations, allowing for a more comprehensive legal assessment. This equation summarizes multiple rounds of retrieval and reasoning, ensuring the final answer reflects the most relevant and accurate information obtained through each iteration.

A key strength of this framework is its feedback loop, which allows for continuous refinement of queries and retrieval of additional context until a high degree of accuracy is achieved. The feedback loop goes beyond the iterative retrieval process, incorporating the results of each retrieval and reasoning stage to inform the next step. For instance, when inconsistencies or gaps in the contract are detected, the feedback loop ensures that the system revisits relevant clauses, further refines its understanding, and adjusts the entire analysis process to address any issues. Unlike single-pass retrieval systems, this iterative process ensures a more comprehensive and precise analysis, particularly in complex legal documents with cross-referenced clauses and embedded dependencies. The feedback loop also helps resolve ambiguities and gaps by retrieving more context and progressively refining the query. This multi-round approach addresses key limitations in existing retrieval systems, making the

proposed framework highly suitable for legal domains requiring detailed and reliable document analysis.

Handling User Queries Post-Analysis. After the initial contract analysis is provided to the user, the system allows for further interaction if the user has specific questions regarding particular sections of the contract. The user can specify a query related to any legal section or compliance check requirement that was not fully addressed or requires more profound insight. Upon receiving the user's query, the system preprocesses the input, generating an initial query for the retrieval process based on the user's request. This stage aims to translate the user's query into a structured format that aligns with the retrieval mechanism, ensuring that the retrieval process captures the intended context of the query.

The Retriever module then identifies the relevant sections of the contract and retrieves them in the same way as in the previous rounds. The LLM Reasoning module compares and analyzes the retrieved clauses, which iteratively refines its understanding based on the query and context provided. The system continues refining the query through multiple rounds, addressing ambiguities, cross-references, and any conflicts between clauses, ensuring that the legal context's response to the user's query is accurate and well-supported. This multi-round retrieval process is essential for providing precise and reliable answers to user queries, especially when the questions relate to complex or interdependent clauses in the contract. Once the system has retrieved and analyzed the necessary clauses, it responds to the user's query, providing actionable insights or clarifications based on the refined understanding of the contract's content. This iterative process ensures that the system can handle user queries in an accurate and contextually relevant way, facilitating an in-depth review of legal documents.

4.2 Prompt Engineering

In designing our method for automated contract reviews, we implemented a structured prompt specifically tailored to align the language model with legal reasoning protocols. The prompt first assigns a legal expert role to the model, then instructs it to analyze the contract clause-by-clause basis, and finally requires the findings to be reported in a consistent JSON format. This layered structure was motivated by the need to reduce the errors and omissions often seen in more open-ended or loosely specified prompts. By providing explicit instructions and a standardized output schema, the prompt ensures that minor clauses, subtle definitions, and attachments receive the same scrutiny as major provisions concerning payment terms or liability. Such thoroughness mitigates the risk of overlooking vital legal nuances—an error that can be costly when translating a contract into real-world obligations.

This approach also addresses the problem of inconsistent or untraceable review results, an issue commonly reported when working with large language models in complex domains. Without predetermined sections for each element (e.g., "clause number," "problem analysis," "revision suggestions"), the model tends to produce free-form text that can be difficult to verify and compare across multiple documents or review cycles. By contrast, the predefined JSON schema not only enforces uniformity—making it easier for legal teams to parse and integrate the output into existing contract management systems—but also serves as a cognitive scaffold for the model. Each required field acts as a prompt for deeper analysis, encouraging more precise reasoning and reducing the likelihood of skipped details. In this sense, the structured prompt does not merely improve output readability; it ensures a full-spectrum legal evaluation, bridging the gap between sophisticated NLP capabilities and the exacting standards of professional contract law.

Table 1: Performance of basic model, single-round RAG, and multi-round RAG configurations on contract analysis.

Method	Retrieval-augmented	Precision	Recall	F1
Baseline(Qwen2.5)	No	44.3	57.3	50.0
Single-round RAG(GPT-4o-mini)	Yes	30.9	38.7	34.3
Multi-round RAG(GPT-4o-mini)	Yes	31.9	40.0	35.5
Single-round RAG(Qwen2.5)	Yes	28.3	74.7	41.0
Multi-round RAG(Qwen2.5)	Yes	28.9	78.7	42.3

4.3 Retrieval and Query Refinement Architecture

The architecture of the retrieval component is designed to handle the unique challenges of legal document review. The architecture combines a semantic embedding model for similarity detection with a multi-round retrieval mechanism. This iterative query refinement ensures accurate and comprehensive retrieval. The retrieval process relies on a pre-trained embedding model (M3E) fine-tuned for Chinese language understanding. The documents are first divided into 512-token chunks to maintain context while optimizing retrieval efficiency. The retrieval engine uses these embeddings to search for relevant clauses across the contract, focusing on semantic matches rather than simple keyword searches. The query refinement model iteratively improves query specificity based on the context of previously retrieved clauses. After each round, the module analyzes and assesses whether retrieved content satisfies the information requirements. If inconsistent, it adjusts the query parameters to target additional related clauses. This process continues until sufficient context is retrieved, ensuring cross-clause consistency and resolving potential ambiguity, resulting in the most accurate response. Suppose new errors or conflicts are identified in the retrieved context. In that case, the model refines its queries based on the legal references retrieved earlier, which helps to ensure that the clauses align correctly with the relevant laws and regulations.

The refinement process terminates once no new relevant information can be found or when the user-depth limit for retrieval rounds is reached. Retrieved clauses are then aggregated and presented alongside explanations, presenting potential inconsistencies, missing cross-references, or conflicting terms.

5 Experiment Setup

The experiments were conducted to evaluate the proposed multi-round retrieval-augmented generation (RAG) framework for legal contract review, focusing on retrieval accuracy. The study aimed to evaluate the system's accuracy in detecting and analyzing contract flaws against several baseline models. A controlled experiment with real-world contracts and systematically modified versions was conducted, while the RAG framework utilized a retrieval dataset of authentic Chinese laws and regulations.

5.1 Datasets

A vast collection of actual laws and regulations that serve as reliable legal references in various fields make up the retrieval dataset that the RAG framework uses. By acting as the retrieval system's fundamental knowledge base, these papers guarantee that the interpretation and analysis of contract clauses are based on the pertinent legal framework. These texts were chunked according to their natural portions, with a maximum of 512 tokens per chunk, to guarantee context retention and effective retrieval. While adhering to the token restriction necessary for transformer-based models, our section-based chunking technique guarantees semantically coherent retrieval. This chunking technique allows the dataset to be processed quickly while preserving a high degree of semantic correctness, essential for the complex legal interpretations required in contract analysis.

This dataset plays a key role in supporting the iterative retrieval process. It allows the system to refine its queries across multiple rounds, using the legal references retrieved to inform the analysis of contract clauses. When errors or ambiguities are detected in the contract, the system queries this dataset to retrieve relevant laws and regulations, which in turn helps the system refine its understanding of the contract's context. This ensures that the legal analysis is based on up-to-date and authoritative sources, making the retrieval results more accurate, reliable, and aligned with legal standards.

The test dataset used for the experiments was constructed from diverse real-world contracts representing various legal and technical domains. These contracts include technical service agreements, data service contracts, cybersecurity consultation agreements, government procurement contracts, and property management agreements. This variety ensures that the dataset covers various contractual structures, clauses, and industry-specific terms, providing a comprehensive testbed for evaluating the proposed framework. The test dataset is crucial for assessing how well the framework performs across different contract types and industries. Five predefined flaws were introduced to create a corresponding modified version for each original contract. These flaws were designed to simulate typical issues found in legal contracts, such as ambiguous deadlines, inconsistent terms, missing cross-references, and altered numerical values. This allows the system to evaluate the model's effectiveness in detecting and addressing various contract-related issues. The flaws were intentionally introduced to create realistic challenges for the framework, testing its ability to detect errors, inconsistencies, and cross-clause conflicts. The resulting dataset comprises standard contracts and their intentionally flawed counterparts, all manually reviewed and labeled to establish a reliable ground truth. This ground truth enables precise evaluation of the system's performance in detecting errors and verifying compliance across different types of contracts. Additionally, the dataset is used to evaluate the framework's ability to handle complex legal scenarios, such as cross-referenced clauses, dependencies, and conflicts between different parts of the contract. By using a labeled dataset with predefined flaws, the evaluation process ensures that the RAG framework's performance can be rigorously tested against a set of known issues, providing an accurate assessment of its ability to detect and resolve typical legal contract problems.

5.2 Evaluation Metrics

We evaluated the system's performance using several standard metrics. Precision measures the proportion of correctly identified flaws among all flagged clauses, while recall quantifies the proportion of flaws successfully detected. The F1 score, the harmonic mean of precision and recall, provides a balanced assessment of the system's overall accuracy. These metrics were selected to comprehensively evaluate the system's effectiveness in identifying and resolving contractual inconsistencies and flaws.

5.3 Baseline and Comparative Models

We compared the multi-round RAG framework's performance with several baseline models to contextualize it. The LLM-only model served as the first baseline, relying solely on the language model without retrieval augmentation. This model provided a benchmark for understanding how much retrieval contributes to overall performance. The second baseline was a single-round RAG model, which performed retrieval in a single pass without query refinement. This model allowed us to quantify the benefits of multi-round refinement in improving retrieval accuracy and context aggregation. These baseline models were chosen to offer diverse comparisons, highlighting the unique contributions of the multi-round RAG framework.

5.4 Implementation Details

The multi-round RAG framework was implemented with Qwen-2.5 (72B) as the primary language model, chosen for its ability to process complex Chinese documents. Retrieval was conducted using M3E embeddings, which provided reliable semantic matching for clause-level context. Contracts were split into 512-token chunks to balance context preservation and retrieval efficiency. In each query round, the system refined its queries based on previously retrieved results, iteratively improving its understanding and retrieval precision. This multi-round approach significantly enhanced the system's ability to detect complex flaws and resolve cross-clause dependencies.

6 Results and Evaluation

6.1 Evaluation

Our systematic evaluation of the proposed framework encompassed five configurations: Basic model (Qwen2.5), Single-Round RAG (GPT-4o-mini and Qwen2.5), and Multi-Round RAG (GPT-4o-mini and Qwen2.5), with results summarized in Table 1. The basic model , which uses a simple large language model without external retrieval, established a foundational performance level with 57.3% recall and 44.3% precision. While the precision is relatively high, this model struggles to identify a wide range of errors, especially those requiring complex reasoning or cross-referencing of clauses.

The reason for the baseline's high precision and recall lies in the fact that it identified a much smaller number of errors compared to the other models. Specifically, of the 75 modified errors in the dataset, the baseline only detected 43 errors. In contrast, the Qwen models in both single and multi-round retrieval configurations were able to detect 56 and 59 errors, respectively. While the baseline identified fewer errors, its high precision and recall metrics were skewed by the fact that it missed a significant number of

Table 2: Comparison of Effect Improvement

Error Category	Single-Round		Multi-Round		Improvement (%)	
	GPT-4o-mini	Qwen2.5	GPT-4o-mini	Qwen2.5	GPT-4o-mini	Qwen2.5
Cross-Clause Dependency	9	**5**	9	**3**	0.0%	**40.0%**
Contractual Imbalance and Unilateral Control	26	**9**	24	**7**	7.7%	**22.2%**
Obligation Scope and Responsibility Errors	11	**7**	12	**7**	-9.1%	**0.0%**

potential issues. This highlights the key trade-off between precision and recall: the baseline model had a low recall because it missed many relevant errors, while the Qwen models had a higher recall, identifying more errors but at the cost of lower precision. The total number of errors identified by the models were as follows: Baseline: 97 errors identified; Qwen Single Round: 198 errors identified; Qwen Multi Round: 204 errors identified.

Thus, although the baseline model reported high precision due to its limited set of detected errors, the Qwen models demonstrated their superiority in identifying a broader set of contract issues. The multi-round Qwen model outperformed the single-round model, as it further refined its understanding of the contract and retrieved additional context, leading to an increase in recall without sacrificing precision too drastically.

The integration of retrieval mechanisms in the single-round RAG using GPT-4o-mini resulted in a recall of 38.7% and a precision of 30.9%. While this recall improvement highlights the benefits of retrieval-augmented generation (RAG), the absolute recall value remained considerably lower than that of Qwen2.5-based retrieval configurations. The multi-round RAG setup with GPT-4o-mini slightly improved recall to 40.0%, reinforcing the effectiveness of iterative retrieval, though it still lagged behind the Qwen2.5-based models in capturing intricate contract details. The limited recall gains suggest that GPT-4o-mini may struggle with complex legal text comprehension compared to Qwen2.5. In contrast, the Qwen2-based configurations yielded substantial improvements in recall. The single-round RAG using Qwen2.5 reached 74.7% recall, a 30.2% relative gain over the baseline. The multi-round RAG configuration with Qwen2.5 further refined this capability, reaching 78.7% recall through iterative context building, particularly excelling in contracts with intricate cross-referenced obligations and nested conditions.

However, this enhanced detection capability introduced an interesting trade-off in precision metrics. Compared to the baseline's 44.3% precision, the single-round RAG (Qwen2.5) model saw a decline to 28.3%, with multi-round RAG (Qwen2.5) showing a slight recovery to 28.9%. A similar pattern was observed with GPT-4o-mini, where single-round RAG resulted in 30.9% precision, and multi-round RAG achieved 31.9%. This pattern reflects the inherent challenge of maintaining precision while expanding the context window for improved recall. Notably, the multi-round configuration's slight precision improvement over single-round suggests that iterative refinement helps filter some irrelevant results, though this remains an area for optimization. The F1-score trends further support these observations. The single-round RAG (GPT-4o-mini) achieved an F1-score of 34.3%, increasing to 35.5% in the multi-round setting. The Qwen2.5-based RAG models demonstrated superior overall effectiveness, with F1-scores of 41.0% for single-round and 42.3% for multi-round, reinforcing their ability to balance retrieval improvements despite the precision trade-offs.

6.2 Error Analysis

We conducted a comprehensive error analysis to evaluate our multi-round RAG framework for contract review. The analysis identified missed errors in three categories (Tabel **??**.), with the multi-round RAG framework showing significant improvement over the single-round configuration, particularly in cross-clause dependency errors. The multi-round approach reduced missed errors by 60% in this category, highlighting the benefits of iterative context refinement. However, challenges remain in handling long-range dependencies in extensive contracts, suggesting the need for enhanced cross-clause tracking mechanisms.

The analysis of contractual imbalance errors, which encompass instances of disproportionate control allocation between parties, revealed a 33% improvement in detection accuracy with the multi-round framework. The single-round configuration missed nine errors, while the multi-round approach reduced this to six. Although the iterative query refinement enhanced the detection of subtle unilateral control instances, the remaining errors highlight limitations in identifying implicit power imbalances that require subjective interpretation. Future improvements may necessitate legal-specific pretraining and the integration of semantic clause relationship models.

Both single-round and multi-round retrieval models demonstrated identical performance in detecting obligation scope errors, with seven missed errors in each configuration. This consistent error rate suggests fundamental limitations in current retrieval strategies regarding the disambiguation of responsibility clauses. The persistent challenge lies in the context-dependent nature of these errors, which often require nuanced interpretation rather than identification of explicit contradictions. Future work should explore the implementation of role-based clause analysis and context-sensitive refinement mechanisms to address these limitations.

7 Conclusion

This study presents a multi-round retrieval-augmented generation framework that enhances legal document analysis by iteratively refining queries and retrieving additional context for improved clause interpretation. By addressing cross-clause dependencies and refining retrieval accuracy, our approach significantly improves the completeness and contextual understanding of contract analysis. Future work will focus on optimizing precision and expanding the framework's applicability to broader legal and regulatory domains.

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